



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 wide notification number F-9-12/2001-U-3 of Government of India)

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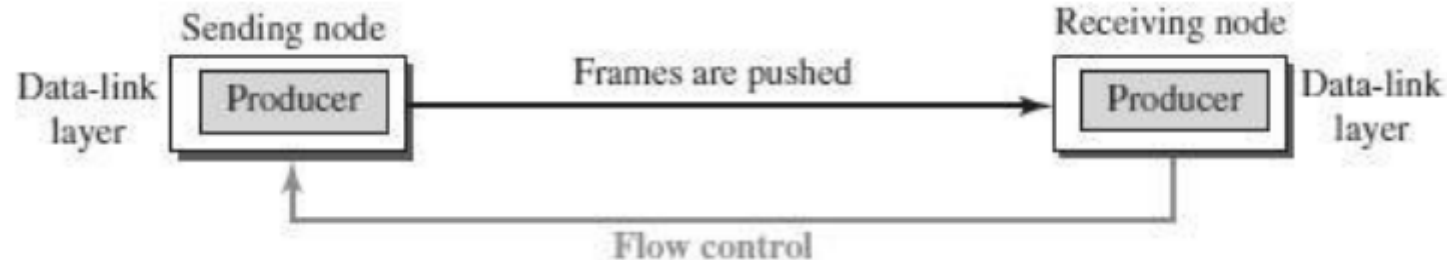
Computer Networks/0705210503

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FLOW CONTROL

- **Flow control refers to a set of procedures used to restrict the amount of data that the sender can send before waiting for acknowledgment.**
- The receiving device has limited speed and limited memory to store the data.
- Therefore, the receiving device must be able to inform the sending device to stop the transmission temporarily before the limits are reached.
- It requires a buffer, a block of memory for storing the information until they are processed.



Two methods have been developed to control the flow of data:

- Stop-and-Wait
- Sliding Window



STOP-AND-WAIT

- The simplest scheme is the stop-and-wait algorithm.
- In the Stop-and-wait method, the sender waits for an acknowledgement after every frame it sends.
- When acknowledgement is received, then only next frame is sent.
- The process of alternately sending and waiting of a frame continues until the sender transmits the EOT (End of transmission) frame.

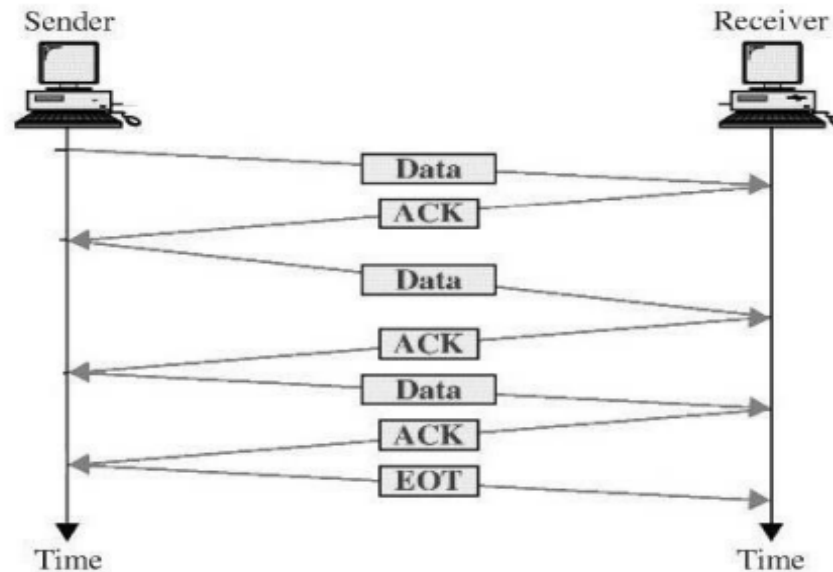


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- If the acknowledgement is not received within the allotted time, then the sender assumes that the frame is lost during the transmission, so it will retransmit the frame.
- The acknowledgement may not arrive because of the following three scenarios :
 1. Original frame is lost
 2. ACK is lost
 3. ACK arrives after the timeout



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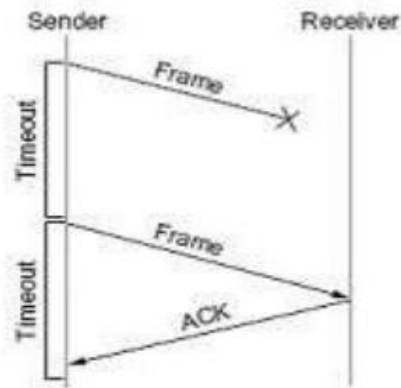
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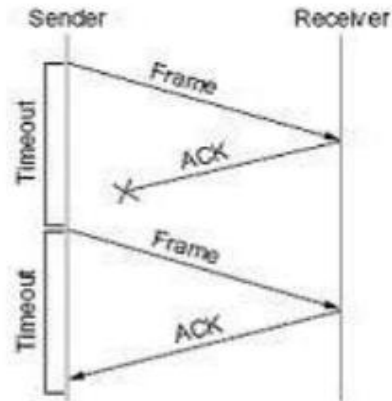
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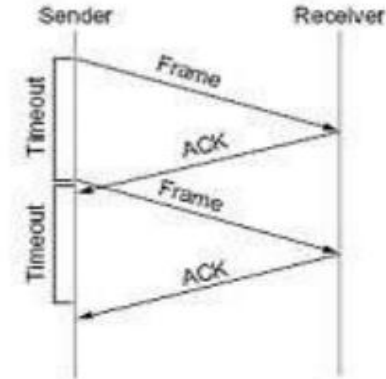
Original frame is lost



ACK is lost



ACK arrives after the timeout



Advantage of Stop-and-wait

- The Stop-and-wait method is simple as each frame is checked and acknowledged before the next frame is sent

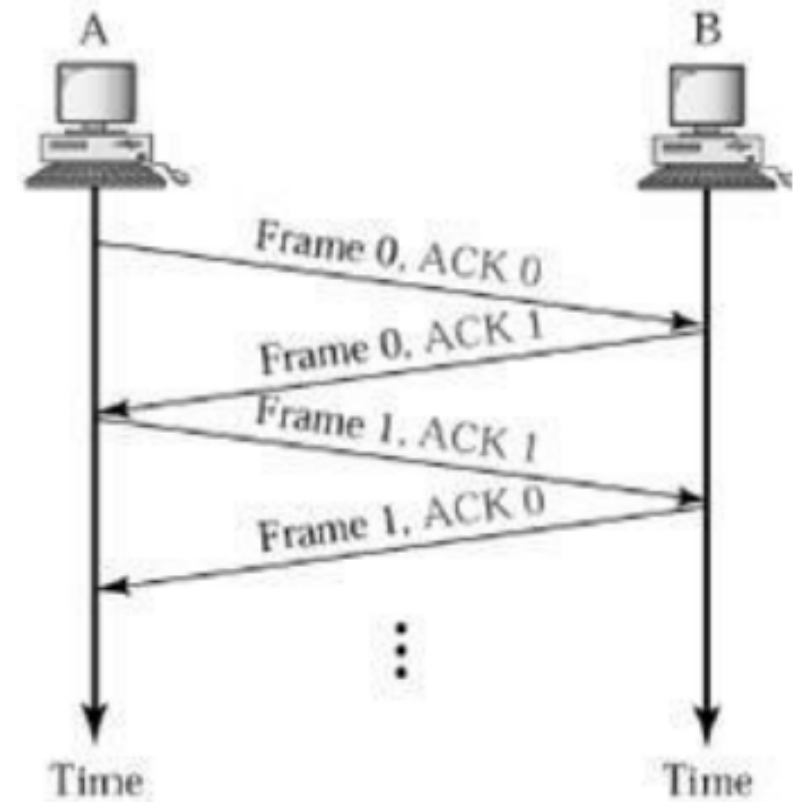
Disadvantages of Stop-And-Wait

- In stop-and-wait, at any point in time, there is only one frame that is sent and waiting to be acknowledged.
- This is not a good use of transmission medium.
- To improve efficiency, multiple frames should be in transition while waiting for ACK.



PIGGYBACKING

- A method to combine a data frame with ACK.
- Piggybacking saves bandwidth
- Station A and B both have data to send.
- Instead of sending separately, station A sends a data frame that includes an ACK.
- Station B does the same thing.





SLIDING WINDOW

- The Sliding Window is a method of flow control in which a sender can transmit the several frames before getting an acknowledgement.
- In Sliding Window Control, multiple frames can be sent one after the another due to which capacity of the communication channel can be utilized efficiently.
- A single ACK acknowledge multiple frames.
- Sliding Window refers to imaginary boxes at both the sender and receiver end.
- The window can hold the frames at either end, and it provides the upper limit on the number of frames that can be transmitted before the acknowledgement.
- Frames can be acknowledged even when the window is not completely filled.
- The window has a specific size in which they are numbered as modulo-n means that they are numbered from 0 to n-1.



- For example, if $n = 8$, the frames are numbered from
0,1,2,3,4,5,6,7,0,1,2,3,4,5,6,7,0,1.....
- The size of the window is represented as $n-1$. Therefore, maximum $n-1$ frames can be sent before acknowledgement.
- When the receiver sends the ACK, it includes the number of the next frame that it wants to receive.
- For example, to acknowledge the string of frames ending with frame number 4, the receiver will send the ACK containing the number 5.
- When the sender sees the ACK with the number 5, it got to know that the frames from 0 through 4 have been received.



Sender Window	Receiver Window
Senderwindow Direction Direction This wall moves to the right When a frame is sent. This wall moves to the right When an ACK is received.	Receiverwindow Direction Direction This wall moves to the right When a frame is received. This wall moves to the right When an ACK is sent.



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| <ul style="list-style-type: none">○ At the beginning of a transmission, the sender window contains $n-1$ frames.○ When a frame is sent, the size of the window shrinks.○ For example, if the size of the window is 'w' and if three frames are sent out, then the number of frames left out in the sender window is $w-3$.○ Once the ACK has arrived, then the sender window expands to the number which will be equal to the number of frames acknowledged by ACK. | <ul style="list-style-type: none">○ At the beginning of transmission, the receiver window does not contain n frames, but it contains $n-1$ spaces for frames.○ When the new frame arrives, the size of the window shrinks.○ For example, the size of the window is w and if three frames are received then the number of spaces available in the window is $(w-3)$.○ Once the acknowledgement is sent, the receiver window expands by the number equal to the number of frames acknowledged. |
|--|---|



Example of Sliding Window

